Jonathan Spraggett

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Github: https://github.com/JonathanSpraggett
Portfolio: https://jonathanspraggett.github.io/

Selected Experience

May 2023 - Current

Robotics Software Engineer — Mostavio

- Lead the design and development of the autonomy stack for a 250 kg MTOW eVTOL designed to carry a 150 kg payload, integrating motion planning, state estimation, and autonomous mission execution using **ROS2**, **C++** and **Python**
- Architected a modular behavior-tree-based autonomy framework, decoupling high-level mission decisions from low-level flight actions to support reusable behaviors, fault handling, and autonomous mission execution.
- Developed a 3D kinodynamic RRT* planner incorporating vehicle dynamics, with minimum-snap trajectory optimization and differential-flatness-based reference generation for smooth, feasible flight.
- Built a constrained 3D model predictive controller (MPC) using quadratic programming for trajectory tracking and obstacle avoidance, integrating NVIDIA nvblox ESDF mapping and velocity, acceleration, and control-rate limits
- Led autonomy, perception, and localization software for Skyguard, an autonomous site-security drone system. Integrated RF-DETR detection, multi-object tracking, camera-to-world target geolocation, and patrol, investigate, watch, and follow behaviors.
- Performed system identification and developed a dynamic model of a coaxial eVTOL to support flight-controller design and tuning, evaluating closed-loop stability through gain and phase margin analysis
- Developed flight-deck and telemetry tools for real-time aircraft monitoring, preflight safety checks, command interfaces and postflight data analysis, enabling repeatable validation of autonomy and flight-control systems.
- Designed and flight-validated digital vibration filters using frequency-domain rotor analysis, improving control performance and reducing tracking RMSE by 60%; published at **AIAA SciTech 2026**.
- Conducted over 1,500 subscale and 100 full-scale eVTOL flight tests, validating flight-control algorithms, autonomous navigation, system stability, and aircraft performance under real-world operating conditions.
- Built Gazebo SITL/HITL simulations and Docker/GitHub Actions CI/CD pipelines, cutting regression-test cycles from hours to under **50** minutes and packaging field deployments as containers.

Sept 2021 - July 2025

Reinforcement & **Imitation Learning Researcher** — University of Toronto Robotics Association

- Developed a unified humanoid recovery policy using CrossQ reinforcement learning in MuJoCo, achieving up to 86% zero-shot recovery success across seven robot morphologies; published at RoboCup Symposium 2025.
- Developed a sim-to-real humanoid locomotion framework using PPO and Adversarial Motion Priors (AMP) in NVIDIA Isaac Gym, leveraging domain randomization to transfer walking, jumping, and kicking policies to physical robot
- Open-sourced a cross-morphology humanoid reinforcement learning benchmark, enabling standardized evaluation of zero-shot policy transfer across different robot platforms.

Aug 2019 - May 2023

Robotic Software Team Lead (Github) — University of Toronto Robotics Association

- Led a 30-member robotics software team developing autonomous humanoid soccer robots, achieving 4th place worldwide at RoboCup 2023.
- Built real-time vision pipelines using YOLO and RF-DETR, reaching 140 FPS on NVIDIA Jetson Orin Nano through model quantization; developed inverse-perspective mapping (IPM) for object localization and tracking.
- Developed ICP-based visual odometry and field-line localization in OpenCV and ROS 2, integrating measurements through an unscented Kalman filter (UKF).
- Developed hierarchical finite-state machine (HFSM) and hierarchical task network (HTN) planners for robot behavior, strategic task decomposition, recovery logic, and temporal dependencies.
- Developed a multi-agent task allocation framework using mixed-integer linear programming (MILP) to optimize robot assignments and task scheduling under resource and timing constraints.

June 2020 - Dec 2022 **Software Developer** Quanta Technology, LLC

- Redesigned graph traversal algorithms for electrical network analysis, reducing computational complexity from $O(n^2)$ to $O(nlogn)$ and execution time by 54%.
- Designed and implemented a scalable SQL database architecture for electrical infrastructure data, supporting ingestion and retrieval of over 50 million records.

2023 Jan - 2023 May **Autonomous Drone for Nuclear Safety Inspections Capstone (Github)** UTIAS

- Designed and developed a cutting-edge drone for nuclear safety inspections. Equipped with CV object detection (YoloV5), the drone can read critical data from radiated zones, while its visual SLAM enables accurate localization, RRT* allows for obstacle-free path planning and precise way-point motion planning for navigation through these zones.
  Supervisor: Professor Jonathan Kelly.

Jul 2021 - Sept 2021 **Machine Learning Researcher (Paper)** University of Toronto

- Developed an entropy-based metric to quantify model-dataset complexity for computer vision models and applied it to track CO_2 emissions from the CV community using Python, TensorFlow, OpenCV, and Numpy.

Publications

Flight-Validated Digital Filtering for Vibration Mitigation on a 250 kg eVTOL
J. Spraggett, N. Pourmostaghim, Y. Sadat-Nejad
AIAA SciTech 2026

Learning to Get Up Across Morphologies: Zero-Shot Recovery with a Unified Humanoid Policy
J. Spraggett
RoboCup Symposium 2025

Sim2Real Reinforcement Learning for Soccer Skills
J. Spraggett
BASc Thesis, University of Toronto, 2023

NoFADE: Analyzing Diminishing Returns on CO_2 Investment
A. Fu, J. Tran, A. Xie, ***J. Spraggett****, et al.*
NeurIPS 2021

Education

April 2023 **Bachelor of Applied Science in Engineering Science** University of Toronto
Major in Robotics with a Minor in Artificial Intelligence
Courses: Mobile Robotics & Perception, Robot Modeling & Control, Probabilistic Reasoning & Learning

Technical Proficiencies

Programming:	C/C++, Python, Rust, Typescript, SQL
Robotics:	ROS 2, PX4, BehaviorTree.CPP, FreeRTOS
Planning & control:	RRT*, MPC/MPPI, minimum-snap optimization, differential flatness, CVXPY/QP, MILP, flight dynamics, system identification
Estimation & perception:	EKF/UKF, visual odometry, sensor fusion, NVIDIA nvblox/ESDF, OpenCV, YOLO, RFDETR, object tracking.
Learning & simulation:	PyTorch, JAX, reinforcement/imitation learning, NVIDIA Isaac Gym/Sim, MuJoCo, Gazebo,SITL/HITL.
Embedded & Hardware:	NVIDIA Jetson, STM32, CAN, PCB Design, CAD

Awards

2025 **GoAERO Stage 1 Winner**(Website) GoAERO

- AeroGuardian: Autonomy-ready rescue eVTOL built to deliver fast, efficient, and accessible emergency response.

Licenses & certifications

2024 **Pilot certificate** - Small Remotely Piloted Aircraft System (RPAS), Visual line-of-sight (VLOS) Transport Canada

Talks & Panels

2025 **UTERC 2025 Alumni Panel** UTERC 2025
2024 **UTRA 2024 Robotics Industry Panel** UTRA